

Prerequisites

- Important**
- Use **Kiro CLI** for this section
 - Complete [Setting Up Kiro](#)

Use Kiro to Understand More About Your AWS Environment

In this workshop, we have pre-deployed a sample application using this [CloudFormation stack](#)

Use Case 1: Create Customizable Inventory Report

Let's use Kiro CLI's AWS tools capability to create a customized inventory report

- In your VS Code Server environment, create a new folder to separate context from other files inside VS Code Server:

```
1 cd /Workshop
2 mkdir inventory
3 cd inventory
```

- Within the VS Code Server environment, start Kiro CLI by running:

```
1 kiro-cli chat
```

- Input a prompt to create your desired inventory report. Feel free to add or remove fields as needed

Important

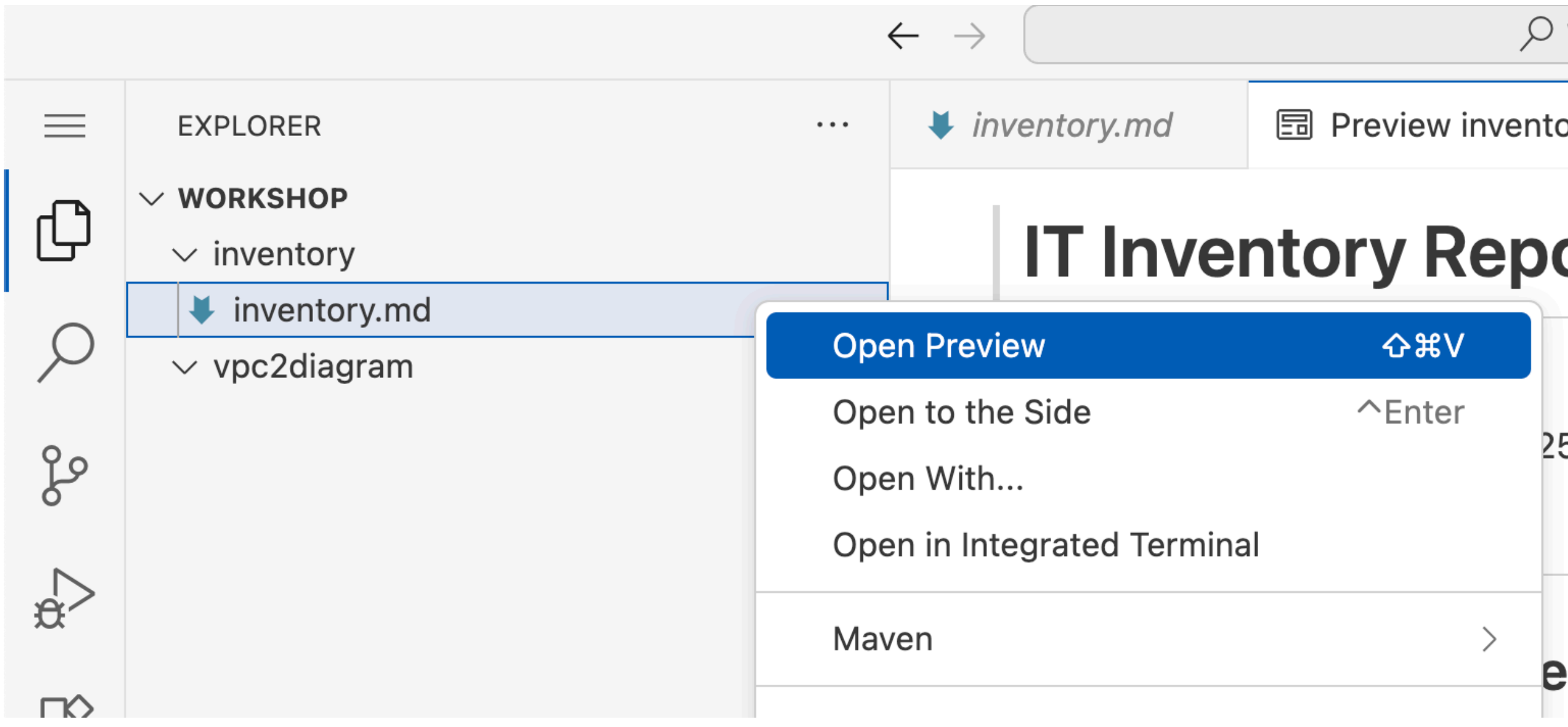
Pro Tip: Kiro CLI follows standard prompt engineering techniques. Context, format, and guidance remain important for optimal results. For multi-line prompts, use the `/editor` command in your Kiro chat session to draft a multi-line prompt. To regenerate, use `/clear` to clear previous thoughts.

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```
1 Create an IT inventory report for server resources in the AWS us-east-1 region. The report should contain:
2 * Resource type
3 * Instance type
4 * vCPU
5 * Memory (GB)
6 * Public IP
7 * Private IP
8 * OS
9 * Open port(s)
10 * IAM role
11 * Comments - based on AWS best practice
12 Summarize the information in to table, which each instance in the same row.
13 Output the report in markdown format save as inventory.md
```

- Open the report by right-clicking the report file > Open Preview



- Sample Report

IT Inventory Report - Server Resources

Region: us-east-1
Report Generated: July 14, 2025
Total Instances: 4

Server Resources Inventory

Resource Type	Instance Type	vCPU	Memory (GB)	Public IP	Private IP	Operating System	Open Ports	IAM Role	Comments (AWS Best Practices)
EC2 Instance (i-0d55bc512ce36cf1d) VSCodeServer	c5.2xlarge	8	16	44.201.137.71	172.31.93.223	Ubuntu 24.04 LTS	80 (HTTP - CloudFront only)	vscode-server-VSCoInstanceProfile-DjpTyK0ADtS	✓ Excellent Security: Uses IMDSv2, CloudFront-only access via prefix list, proper IAM role, monitoring enabled. Follows AWS security best practices.
EC2 Instance (i-01f7517c6dd2d2856) Web Server #1	t3.micro	2	1	None	10.0.4.254	Amazon Linux 2023	22 (SSH), 80 (HTTP) Both open to 0.0.0.0/0	sample-app-EC2InstanceProfile-ktoMh9q46pCW	⚠ Security Risk: SSH and HTTP open to internet. Recommendation: Use AWS Systems Manager Session Manager for SSH, implement ALB for HTTP traffic.
EC2 Instance (i-0dc8137f5dfbaa9cd) Load Generator	t3.micro	2	1	44.198.58.230	10.0.1.105	Amazon Linux 2023	22 (SSH), 80 (HTTP) Restricted to 49.49.242.24/32	None	✓ Good Security: Proper IP restriction. ⚠ Missing IAM Role: Attach IAM role following least privilege principle for AWS service integration.
EC2 Instance (i-03ec69c8db2ec7d19e) Web Server #2	t3.micro	2	1	None	10.0.3.86	Amazon Linux 2023	22 (SSH), 80 (HTTP) Both open to 0.0.0.0/0	sample-app-EC2InstanceProfile-ktoMh9q46pCW	⚠ Security Risk: SSH and HTTP open to internet. Recommendation: Use AWS Systems Manager Session Manager for SSH, implement ALB for HTTP traffic.

- To exit Kiro CLI (when you start a new session, the context in the session will be cleared automatically)

```
1 /quit
```

Use Case 2: Create a Diagram from Existing VPC

Let's use Kiro CLI's AWS tools capability to create an architecture diagram based on resources in your VPC

- In your VS Code Server environment, create a new folder to separate context from other files inside VS Code Server:

```
1 cd /Workshop
2 mkdir vpc2diagram
3 cd vpc2diagram
```

- Within the VS Code Server environment, start Kiro CLI by running:

```
1 kiro-cli chat
```

- Input a prompt to create a diagram from the deployed VPC

```
1 Generate a comprehensive AWS VPC architecture diagram for VPC name 'sampleapp-VPC' in us-east-1. Include:
2 * VPC and Subnets with CIDR
3 * Internet gateways
4 * NAT gateways
5 * Instances in VPC with IP information
6 * Do not need to provide resource association
7 Output the diagram in draw.io format.
8 Please use AWS service icons
9 Save as vpc-diagram.drawio
```

(Optional) You can also use this prompt in your environment by changing the VPC ID and AWS Region

Important

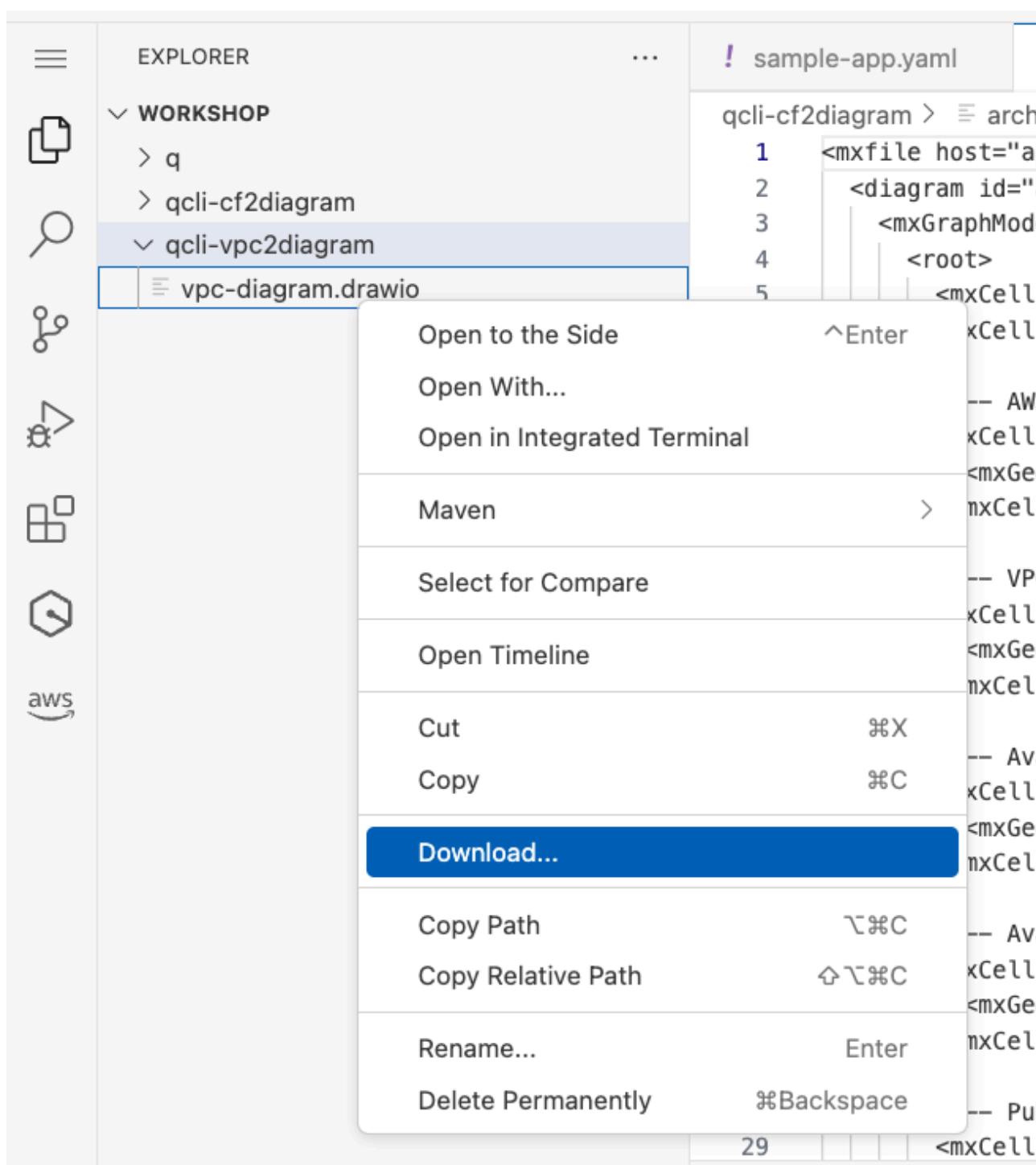
During operation, Kiro may ask for user permission to perform non-read-only actions, such as creating new files or running shell scripts. Users must approve these actions by typing "y" (yes) or "t" (trust). If an action is trusted, Kiro will not ask for permission for the same action in future operations.

Allow this action? Use 't' to trust (always allow) this tool for the session. [y/n/t]:

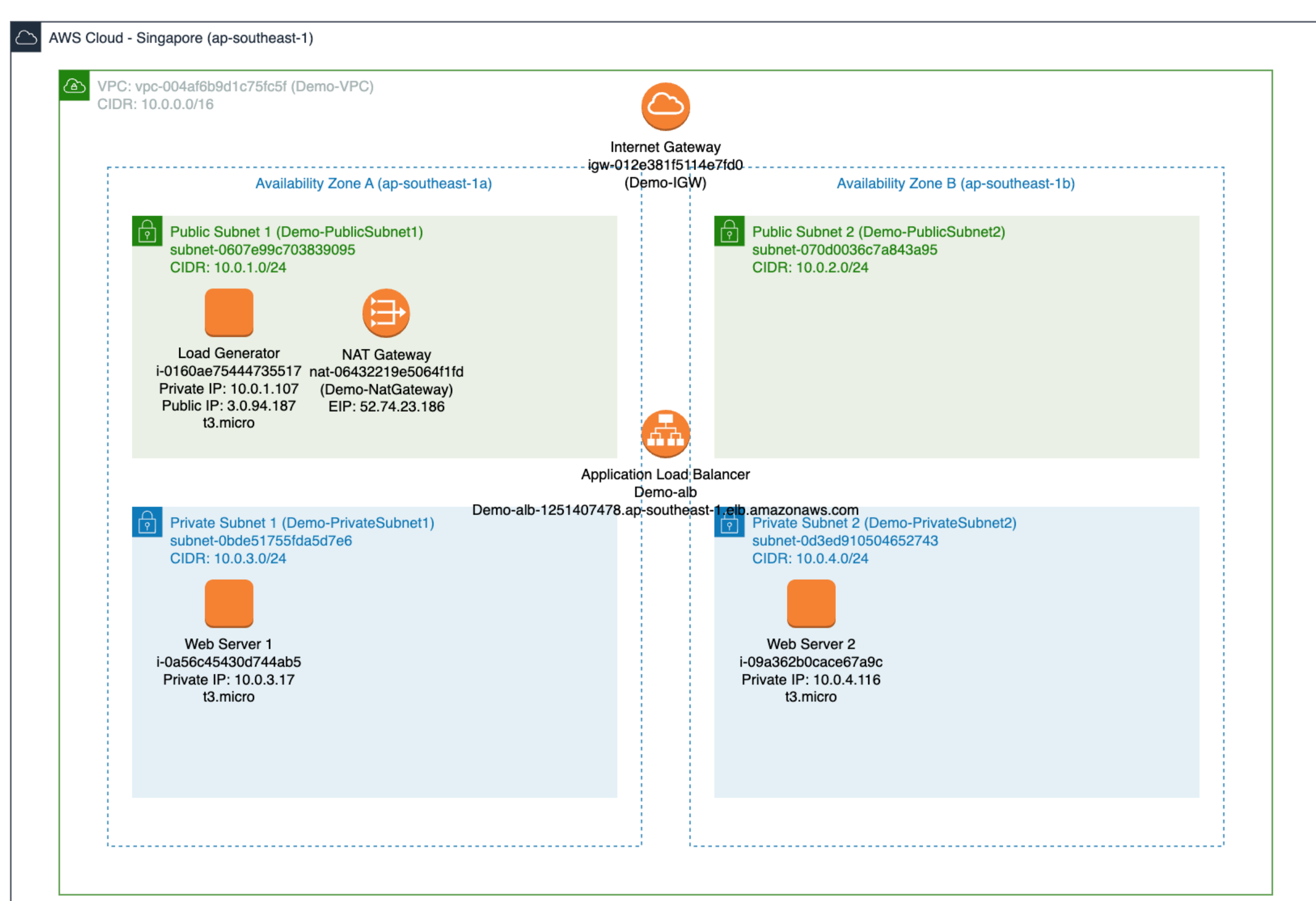
> t

- Kiro will try to gather information about the VPC using AWS CLI and create a diagram based on what's in your VPC. The output file will be placed in the directory where you started Kiro CLI

- Download the diagram to your local machine



- Open the draw.io file at [Draw.io](#) where you should see an architecture diagram that looks like this. Due to the non-deterministic behavior of LLMs, the diagram may not be exactly the same as shown below



- To exit Kiro CLI (when you start a new session, the context in the session will be cleared automatically)

```
1 /quit
```